

B decreases further from wire A and radius increase due to decreasing F . B1

- 7 (a) (i) EM produced whenever charged particle is suddenly accelerated/ decelerated at the metal target (and wavelength depends on magnitude of acceleration) M1
 Electrons hitting the metal target have a range/distribution of accelerations A1
- (ii) All kinetic energy of one electron given up in one collision to produce a single X-ray photon. B1
 Minimum wavelength for maximum energy Or $\lambda_{\min} = hc/E_{\max}$ B1
 So independent of target metals (only depend on accelerating voltage) A0
- (b) (i) More likely (higher probability) for electrons at the next higher level to drop down to fill up the hole, so higher intensity for K_{α} A1
- (ii) At low voltages, the energy of electrons (25 keV) is not sufficient B1
 to knock electrons out of the inner shells of the tungsten atom B1
 So no characteristic X-rays produced by de-excitation. A0
- (c) $E = \frac{hc}{e\lambda} = \frac{(6.63 \times 10^{-34})(3 \times 10^8)}{(1.6 \times 10^{-19})(6.6 \times 10^{-11})}$ M1
 $= 1.88 \times 10^4 \text{ eV}$ A1

- 8 (c) It is the number of trains arriving at a station per unit time. A1